MILITARY SPECIFICATION SHEET

INTERCOMMUNICATION SETS, AN/AIC-14 AND AN/AIC-14A

DUAL OUTPUT AMPLIFIER MODULE

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope.

This specification sheet provides performance and physical characteristics for the Dual Output Amplifier used in Audio Frequency Amplifier AM-3889/AIC-14A.

1.2 Military Part Number.

The military part number shall be M22352/8 -1

2. APPLICABLE DOCUMENTS

Refer to 2.2 of MIL-I-22352(AS) for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or reference document.

3. REQUIREMENTS

3.1 Item Definition. The Dual Output Amplifier module shall provide two amplifiers having a high degree of electrical isolation between the two inputs and between the two outputs.

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The amplifier shall provide amplification of audio signals with gain characteristics as specified herein when used in Audio Frequency Amplifier AM-3889/AIC-14A.

3.1.1 Item Diagram. The Dual Output Amplifier shall provide the functions represented schematically in accordance with the diagram of Figure 1.

3.1.2 Interface Definition. The Dual Output Amplifier input and output signals will normally display the characteristics presented in Table I.

TABLE I. Signal characteristics

Signal Name	Characteristics
Inputs:	
Audio No. 1 (AUDIO NO. 1)	Voice range audio (300 to 7,000 Hz) signal ranging in amplitude from 0 to 14.0 Vac. No dc component normally present.
Inputs:	
Audio No. 2 (AUDIO NO. 2)	Voice range audio (300 to 7,000 Hz) signal ranging in amplitude from 0 to 14.0 Vac. No dc component normally present.
Power (PWR)	+27.5 ±0.5 Vdc
Outputs:	
600 Ohm No. 1 (600 OHM NO. 1)	Voice and sidetone range audio (300 to 7,000 Hz) signal ranging from 0 to 13.4 Vac. Signal power range is 0 to 300 mW.
600 Ohm No. 2 (600 OHM NO. 2)	Voice and sidetone range audio (300 to 7,000 Hz) signal ranging from 0 to 13.4 Vac. Signal power range is 0 to 300 mW.

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TABLE I. Signal characteristics continued.

Signal Name	Characteristics
7.5 Ohm No. 1 (7.5 OHM NO. 1)	Voice and sidetone range audio (300 to 7,000 Hz) signal ranging from 0 to 3.87 Vac. Signal power range is 0 to 2 W.
7.5 Ohm No. 2 (7.5 OHM NO. 2)	Voice and sidetone range audio (300 to 7,000 Hz) signal ranging from 0 to 3.87 Vac. Signal power range is 0 to 2 W.
Power (PWR GND)	Power return. See 3.4.4.2 of MIL-I-22352(AS).
System (SYS GND)	Signal return. See 3.4.4.3 of MIL-I-22352(AS).

3.2 Characteristics

3.2.1 Performance Characteristics

3.2.1.1 Input Characteristics

3.2.1.1.1 Audio No. 1 (AUDIO NO. 1). The Dual Output Amplifier shall have an AUDIO NO. 1 input circuit having the following performance characteristics:

3.2.1.1.1.1 Signal Frequency. The amplifier shall accept AUDIO NO. 1 input drive signals which range from 300 to 7,000 Hz.

3.2.1.1.1.2 Signal Amplitude. The amplifier shall accept AUDIO NO. 1 input drive signals which range in amplitude from 1.2 to 14.0 Vac. The amplifier shall be capable of withstanding overvoltage inputs of up to 16.0 without sustaining damage.

3.2.1.1.1.3 Input Impedance. The amplifier shall have input impedance of 2400 ohms ±10 percent.

3.2.1.1.1.4 Isolation. The isolation between AUDIO NO. 1 and AUDIO NO. 2 signal inputs shall be not less than 20 dB.

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3.2.1.1.2 Audio No. 2 (AUDIO NO. 2). The AUDIO NO. 2 input characteristics shall be as described in Section 3.2.1.1.1 for AUDIO NO. 1.

3.2.1.1.3 Power (PWR). The amplifier shall accept and operate within the constraints of this specification when supplied with +27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1. The maximum load presented by the amplifier shall not exceed 400 milliamperes.

3.2.1.2 Output Characteristics. The amplifier shall have the following output characteristics:

3.2.1.2.1 600 ohm No. 1 (600 OHM NO. 1). The 600 OHM NO. 1 output shall have the following characteristics:

3.2.1.2.1.1 Frequency Response. The amplifier shall have unity gain. The output shall be within ±2 dB of the input amplitude at 1,000 Hz. Within the frequency range from 300 to 7,000 Hz, the signal amplitude shall be flat within ±2 dB of the reference value at 1000 Hz when terminated in a 600 ohm resistive load.

3.2.1.2.1.2 Signal Amplitude. The amplifier shall have unity gain. The output shall equal a nominal input of 0.06 V at 1,000 Hz applied to the AUDIO NO. 1 input within ±2 dB when terminated in a 600 ohm load. The input/output characteristics (gain) shall be linear within ±1 dB for input levels between 0.01 and 1.5 Volts rms and linear within ±3 dB for input levels between 1.5 and 3.0 Volts rms.

3.2.1.2.1.3 Distortion. When operating at full rated power output, the total harmonic distortion over the frequency range of 300 to 7,000 Hz shall not exceed 5 percent at standard conditions as defined in 3.1a of MIL-STD-810. The total harmonic distortion over the frequency range from 300 to 7,000 Hz shall not exceed 10 percent over the entire environmental envelope of MIL-E-5400 Class 2.

3.2.1.2.1.4 Output Power. The output of the amplifier shall be capable of delivering 300 milliwatts into a 600 ohm resistive load.

3.2.1.2.1.5 Output Stability. The output shall show no evidence of regeneration or oscillation throughout the environmental envelope of MIL-E-5400 Class 2 under any combination of input-output open, short, or complex impedance terminations. The Dual Output Amplifier shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specification when properly terminated.

3.2.1.2.1.6 Internal Noise. The Dual Output Amplifier shall have a 600 OHM NO. 1 output amplitude of less than 5 millivolts when operated under standard conditions as defined in 3.1a of MIL-STD-810 with the input circuit shorted to system ground.

3.2.1.2.1.7 Susceptibility to Stray Magnetic Fields. The amplifier shall be shielded or so designed that when oriented in any position within a

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magnetic field of 2 Gauss in the frequency range of 400 to 2,800 Hz, the output from the amplifier shall not be greater than 5 millivolts at 400 Hz. Between 400 and 2,800 Hz the output shall not exceed a linear increase up to 50 millivolts at 2,800 Hz.

3.2.1.2.1.8 Output Impedance. The 600 OHM NO. 1 output shall be capable of delivering rated power when terminated in a 600 ohm resistive load.

3.2.1.2.1.9 Isolation. When the input circuit is properly terminated but no signal is applied, any signal appearing on this outgoing line shall be considered as undesired signal. The undesired signal level on this outgoing line shall be at least 60 dB below rated power output for a normal signal. Undesired signal level on this outgoing line shall be measured with the amplifier properly terminated and the other channel driven to rated power output.

3.2.1.2.2 600 ohm No. 2 (600 OHM NO. 2). The 600 OHM NO. 2 output shall have characteristics identical to those described in 3.2.1.2.1 and subparagraphs there of for the 600 OHM NO. 1 output.

3.2.1.2.3 7.5 ohm No. 1 (7.5 OHM NO. 1). The 7.5 OHM NO. 1 output shall have the following characteristics:

3.2.1.2.3.1 Frequency Response. Within the frequency range from 300 to 7,000 Hz the signal amplitude shall be flat within ±1 dB when terminated in a 7.5 ohm resistive load.

3.2.1.2.3.2 Signal Amplitude. The 7.5 OHM NO. 1 output power shall exceed a nominal input of 0.075 W at 1,000 Hz applied to AUDIO input by 14 ±2 dB when terminated in a 7.5 Ohm load. The input/output characteristics (gain) shall be linear within ±1 dB for input levels between 3.0 and 12.0 Volts rms. The input/output characteristic (gain) shall be linear within ±3 dB for input levels between 1.5 and 3.0 Volts rms.

3.2.1.2.3.3 Distortion. The total harmonic distortion over the frequency range of 300 to 7,000 Hz shall not exceed the values listed in Table II below.

TABLE II. Maximum harmonic distortion.

Condition	Output Level (watts)	Distortion Limit (percent)
Standard	1	5
All Environmental	1	10
Standard	2	10
All Environmental	2	15

Where:
Standard - Standard conditions as defined in 3.5 of MIL-I-5422.
All - The entire environmental envelope defined in MIL-E-5400 for Class 2 equipment.

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3.2.1.2.3.4 Output Power. The 7.5 OHM NO. 1 output of the amplifier shall be capable of delivering 2 watts into a 7.5 Ohm resistive load.

3.2.1.2.3.5 Output Stability. The output shall show no evidence of regeneration or oscillation throughout the environmental envelope specified in MIL-I-22352(AS) under any combination of input-output open, short, or complex impedence terminations. The Dual Output Amplifier shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specification when properly terminated.

3.2.1.2.3.6 Internal Noise. The amplifier shall have an output of less than 5 millivolts when operated under standard conditions as defined in MIL-T-5422 with the input circuit shorted to system ground.

3.2.1.2.3.7 Susceptibility to Stray Magnetic Fields. The Dual Output Amplifier shall be shielded or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 to 2,800 Hz, the output from the Dual Output Amplifier shall not be greater than 5 millivolts at 400 Hz. Between 400 and 2,800 Hz the output shall not exceed a linear increase up to 50 millivolts at 2,800 Hz.

3.2.1.2.3.8 Output Impedance. The 7.5 OHM NO. 1 output shall be capable of delivering rated power when terminated in a 7.5 Ohm ±10 percent resistive load.

3.2.1.2.3.9 Isolation. When the input circuit is properly terminated but no signal is applied, any signal appearing on the outgoing line shall be considered as undesired signal. The undesired signal level on the outgoing line shall be at least 60 dB below rated power output for a normal signal. Undesired signal level on this outgoing line shall be measured with the amplifier properly terminated and the other channel driven to rated power output.

3.2.1.2.4 7.5 ohm No. 2 (7.5 OHM NO. 2). The 7.5 OHM NO. 2 output shall have characteristics identical to those described in 3.2.1.2.3 and subparagraphs there of for the 7.0 OHM NO. 2 output.

3.2.2 Physical Characteristics

3.2.2.1 Dimensions. The amplifier shall consist of a signal encapsulated module having the dimensions of Figure 2. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The amplifier shall weigh no more than 14 ounces.

3.3 Design and Construction. The design and construction of the amplifier shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

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4. QUALITY ASSURANCE PROVISIONS

See MIL-I-22352(AS).

4.1 Responsibility for Tests and Inspections. See MIL-I-22352(AS).

4.2 Test Equipment. See 4.3.3.1 of MIL-I-22352(AS).

4.3 Test Methods.

4.3.1 Examination of Product. See 4.3.3.2.1 of MIL-I-22352(AS).

4.3.2 Operational Test. See 4.3.3.2.2 of MIL-I-22352(AS).

4.3.2.1 Input Impedance. Verify that the AUDIO NO. 1 and AUDIO NO. 3 input impedances are in accordance with 3.2.1.1.1.3 herein.

4.3.2.2 Gain. Verify that the amplifier gain at 1000 Hz is in accordance with 3.2.1.2.1.2 herein.

4.3.2.3 Frequency Response. Holding the input signal constant vary the frequency between 300 Hz and 7000 Hz. Verify that the amplifier gain meets the requirements of 3.2.1.2.1.1 herein.

4.3.2.4 Linearity. Using a 1000 Hz input signal which is varied between 0.01 Vac and 1.5 Vac, verify that the amplifier meets the requirements of 3.2.1.2.1.2.

4.3.2.5 Output Power. With the amplifier property terminated, apply a 1000 Hz signal of appropriate amplitude to verify the requirements of 3.2.1.2.1.4 and 3.2.1.2.3.4 herein.

4.3.2.6 Distortion. Verify the requirements of 3.2.1.2.1.3 herein.

4.3.2.7 Amplifier Stability. Verify the requirements of 3.2.1.2.1.5 herein.

4.3.2.8 Internal Noise. Verify the requirements of 3.2.1.2.1.6 herein.

4.3.2.9 Susceptibility to Stray Magnetic Fields. Verify the performance of 3.2.1.2.1.7 herein.

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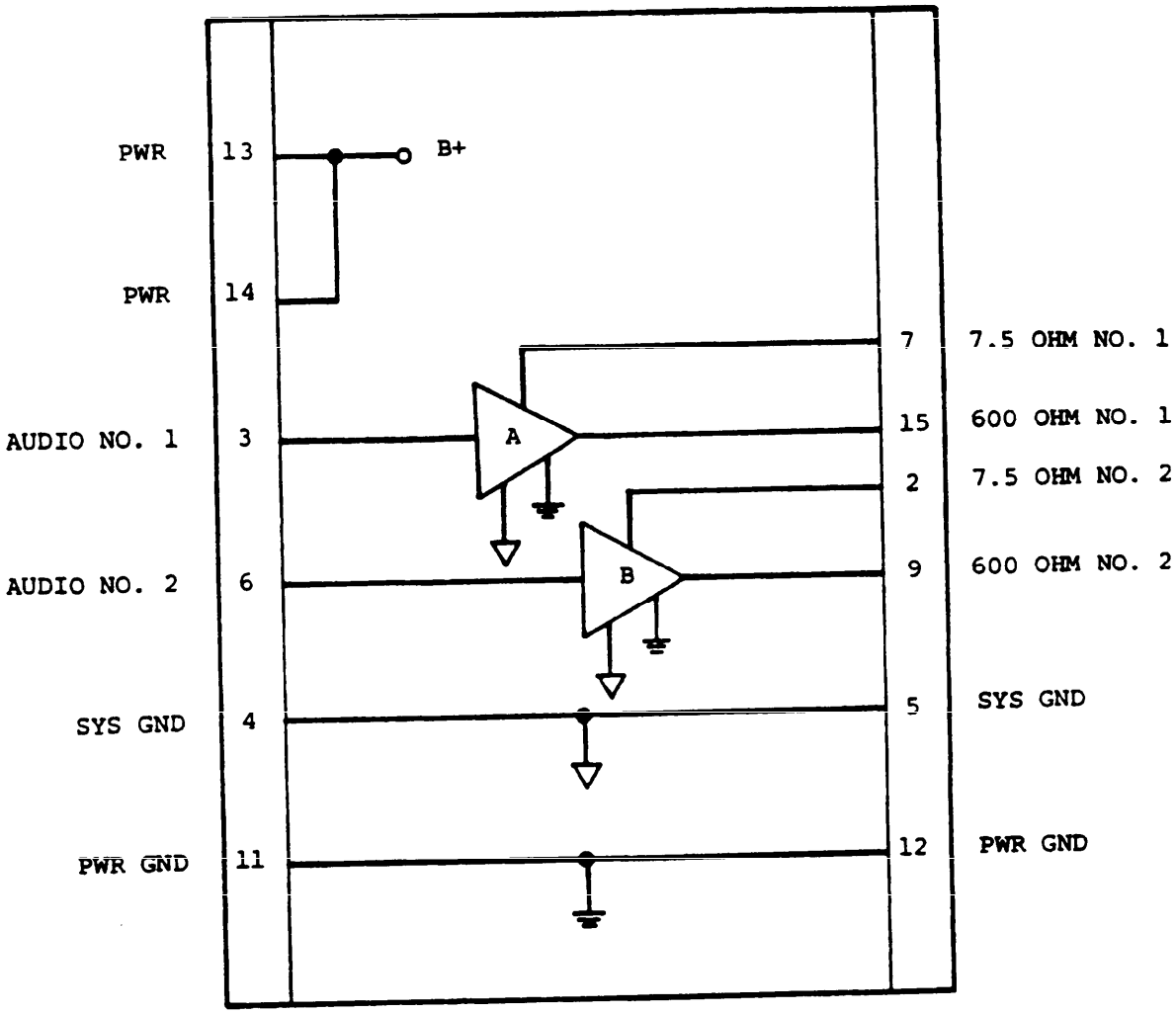
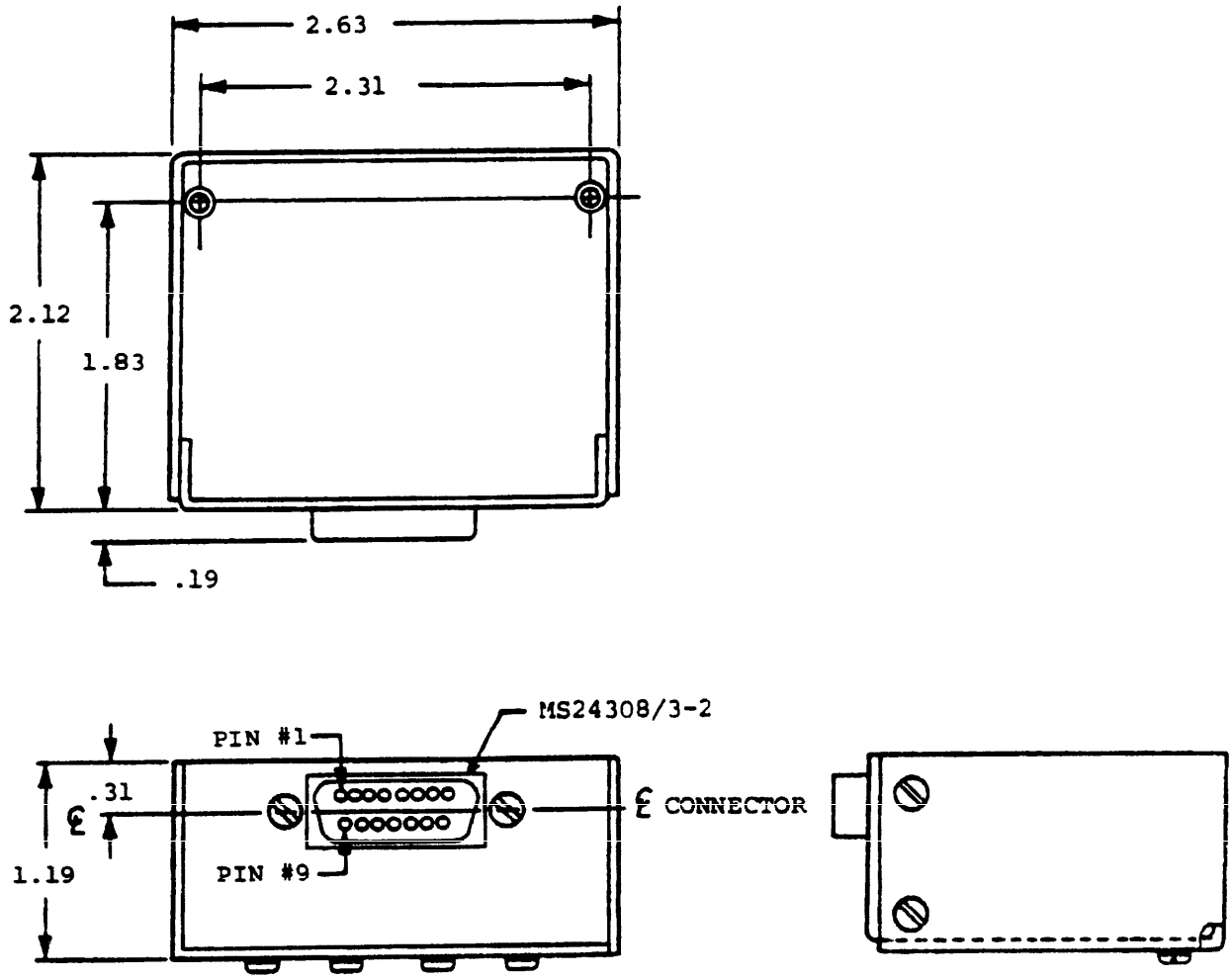


FIGURE 1. Dual output amplifier functional diagram.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 2. Dual output amplifier module outline drawing.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22352/8(AS)	INTERCOMMUNICATION SETS, AN/AIC-14 AND AN/AIC-14A DUAL OUTPUT AMPLIFIER MODULE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5 PROBLEM AREAS

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b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

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